

AAA — Authentication, Authorization & Accounting Assignment

Question 1

Create a table comparing Authentication, Authorization, and Accounting (AAA) by listing at least two real-world examples of each function from popular apps like Instagram, Zomato, or Paytm.

Answer

AAA stands for **Authentication, Authorization, and Accounting**. These three functions help applications verify users, control what they can do, and maintain records of important activities.

AAA Function	Meaning	Real-World Example 1	Real-World Example 2
 Authentication	Verifies the identity of a user	Instagram: Login using username/password or other verification methods	Paytm: User verifies identity when signing in or performing sensitive actions
 Authorization	Determines what an authenticated user is allowed to do	Zomato: A customer can place orders, while a restaurant account manages its restaurant information	Instagram: An ordinary user can manage their own account, while platform administrators have additional management privileges
 Accounting	Records important user/session/activity information	Paytm: Transaction records such as amount, time, and transaction status	Zomato: Order records such as order time, restaurant, order status, and payment status

★ Simple Way to Remember

Authentication → “Who are you?”

Authorization → “What can you do?”

Accounting → “What happened?”

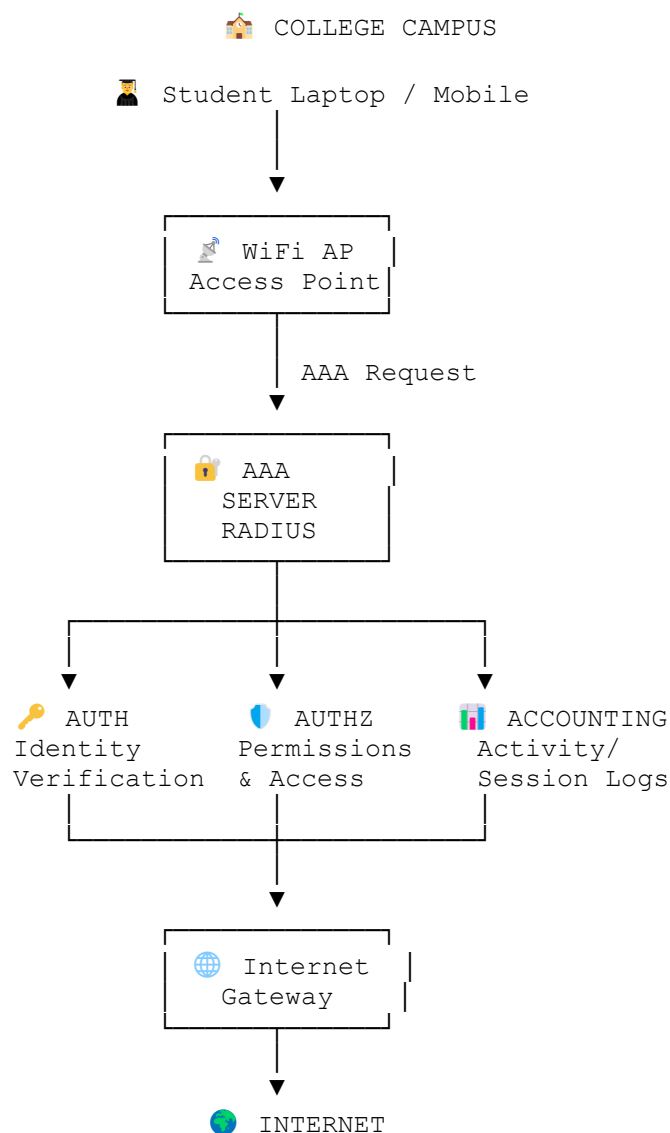
Question 2

Draw a simple network diagram showing how AAA is used to secure access to a WiFi network in a college campus, labeling where authentication, authorization, and accounting take place.

Hint: Include devices like user laptops, access points, AAA server, and the internet gateway.

Answer

🏠 College WiFi — AAA Network Diagram



🔑 Authentication

The student enters their college credentials. The WiFi access point forwards the authentication request to the **AAA/RADIUS server**, which verifies whether the student's identity is valid.

Authorization

After successful authentication, the AAA system determines what the student is permitted to access. For example, students may receive Internet and educational-resource access while staff or administration networks remain restricted.

Accounting

The AAA system records relevant session information such as **username, login time, logout time, assigned IP address, and network-session information**. These records can assist with monitoring and security investigations.

Question 3

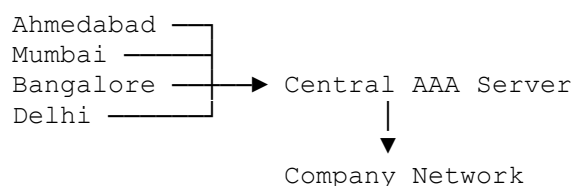
Explain the difference between centralized and distributed AAA deployment models by describing how each would work in a scenario where multiple Flipkart warehouses need secure network access.

Answer

Consider Flipkart operating warehouses in **Ahmedabad, Mumbai, Bangalore, and Delhi**. Employees at each warehouse need secure access to company applications and internal systems.

Centralized AAA

In a **centralized AAA model**, the warehouses use one or more centrally managed AAA servers located in a central data center or cloud environment.



How it works

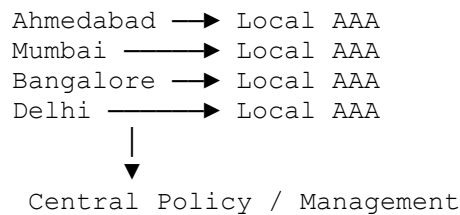
1. An employee at the Ahmedabad warehouse attempts to access the corporate network.
2. The access device sends the authentication request to the centralized AAA server.
3. The AAA server verifies the employee and applies the appropriate access policy.
4. Similar requests from the other warehouses are handled by the centralized AAA infrastructure.

Main characteristic

One centrally managed AAA infrastructure serves multiple locations.

Distributed AAA

In a **distributed AAA model**, AAA services are deployed across multiple locations or regions.



How it works

Each warehouse can have access to a **local or regional AAA server**. Employees can authenticate against a nearby AAA service, while policies and identity information can be coordinated across the organization's infrastructure.

Main characteristic

AAA services are distributed across multiple locations instead of relying entirely on one central service.

Comparison

Feature	Centralized AAA	Distributed AAA
AAA location	Central infrastructure	Multiple locations
Management	Easier to centralize	More complex
Availability	Depends heavily on connectivity to central infrastructure	Can continue locally if connectivity to another location is disrupted
Scalability	Can require scaling central infrastructure	Can scale by adding regional infrastructure
Policy management	Centralized	Requires policy synchronization/coordination

Question 4

Pick any one AAA deployment model and list three advantages and two disadvantages of using it for a large-scale app like BookMyShow.

Answer

Selected Model: Centralized AAA

Consider BookMyShow operating services used by customers across many cities.

Advantages

1. Centralized User Management

User authentication policies can be managed from a central infrastructure, making it easier to maintain consistent access controls.

2. Consistent Security Policies

The organization can apply common authentication and authorization policies across its applications and services.

3. Centralized Logging and Monitoring

Authentication and authorization events can be collected centrally, making security monitoring and investigation easier.

Disadvantages

1. Dependency on Central Infrastructure

If the centralized AAA infrastructure becomes unavailable and there is no redundancy or fallback mechanism, authentication services could be affected.

2. Scalability Requirements

A large application may have a very high number of authentication requests. The centralized AAA infrastructure therefore needs appropriate scaling, redundancy, and load distribution.

Practical Approach

For a large-scale service, centralized policy management can be combined with **redundant and geographically distributed AAA infrastructure** to avoid relying on a single server or location.

Question 5

Use ChatGPT or another AI tool to generate a sample AAA policy for a Zomato-like app, specifying what actions are authenticated, which roles are authorized for each action, and what accounting logs should be kept. Paste the generated policy and highlight one thing you would change to make it more secure.

Answer

Sample AAA Security Policy — Food Delivery Application

Policy Name: Food Delivery Application AAA Policy

Purpose:

To ensure that users are properly authenticated, only authorized users can perform specific actions, and important security and transaction events are recorded.

1. Authentication Policy

The following actions require user authentication:

- User login
- Viewing personal account information
- Placing an order
- Making a payment
- Viewing order history
- Restaurant account management
- Delivery-partner account access
- Administrator access

Sensitive operations should require **additional verification** where appropriate.

2. Authorization Policy

Role	Authorized Actions
 Customer	Browse restaurants, place orders, view own orders, manage own profile
 Restaurant Staff	Manage restaurant menu, view assigned orders, update order status
 Delivery Partner	View assigned deliveries and update delivery status
 Administrator	Manage users, restaurants, delivery accounts, and security policies
 Security Administrator	Manage security settings, access policies, and audit systems

Users must not be allowed to perform actions outside the permissions associated with their assigned role.

3. Accounting / Logging Policy

The application should maintain appropriate security and audit records for events such as:

- Successful and failed login attempts
- Password or authentication-factor changes
- Account creation and account changes
- Order creation and cancellation
- Payment events and transaction status
- Changes to restaurant or delivery information
- Administrator actions
- Permission/role changes
- Important security events

Logs should include relevant information such as:

Timestamp + User/Account ID + Event Type + Result + Source Information

Sensitive information such as passwords and authentication secrets **must not be stored in plain text in logs.**

One Security Improvement I Would Make

★ Add Multi-Factor Authentication (MFA) for Sensitive Accounts

I would require **MFA for administrator, restaurant-management, and other privileged accounts**, rather than relying only on a username and password.

For example:

```
Username + Password
      +
Second Authentication Factor
      ↓
Access Granted
```

This provides an additional security layer if a password is compromised.

Final Revision

AAA

Main Question

Food-Delivery Example



Authentication Who are you?

Customer logs into their account



Authorization What can you do? Restaurant staff can manage their own menu



Accounting What happened? Login, order, payment, and admin events are logged

★ Key Takeaway

AAA provides three essential security controls: Authentication verifies identity, Authorization controls permissions, and Accounting records important activities for monitoring and auditing.